Australian Government

Department of the Environment and Energy

Cloud Strategy

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DEPARTMENT OF THE ENVIRONMENT AND ENERGY

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ACKNOWLEDGEMENT OF COUNTRY

The Department acknowledges the traditional owners of country throughout Australia and their continuing connection to land, sea and community. We pay our respects to them and their cultures and to their elders both past and present.

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REFERENCES

Serial	Title	Source and Version
A.	DTA Secure Cloud Strategy	https://www.dta.gov.au/our-projects/secure-cloud-strategy
B.	ASD Certified Cloud Services	https://acsc.gov.au/infosec/irap/certified_clouds.htm

Table 1 – References

1 EXECUTIVE SUMMARY

'The cloud' means storing and accessing data and programs by way of the Internet. Cloud services deliver on-demand computing resources over the Internet as an alternative to maintaining Department owned physical information and communication technology (ICT) infrastructure, corporate data centres, platforms and software.

Organisations are increasingly looking to cloud services to help meet their changing business demands. Industry experience has shown that when done well, cloud services can provide organisations innovative ways to deliver existing services, whilst also providing rapid access to new capability. When adopted in a planned and well-governed manner, cloud services can also help reduce ICT operational costs.

The Department of the Environment and Energy recognises cloud services as key to supporting the delivery of business outcomes, now and into the future. The benefits of cloud adoption for the Department include the enablement of a more flexible and modern workplace, faster delivery of new Government programs and initiatives and enhanced cross agency and community engagement. Adoption of cloud services will allow the Department to leverage innovative new technologies without the need to sustain specialist skills in house.

The Department's business, technology and Whole of Government drivers for adoption of cloud services are discussed in section 3 of this document and summarised in **Figure 1**.

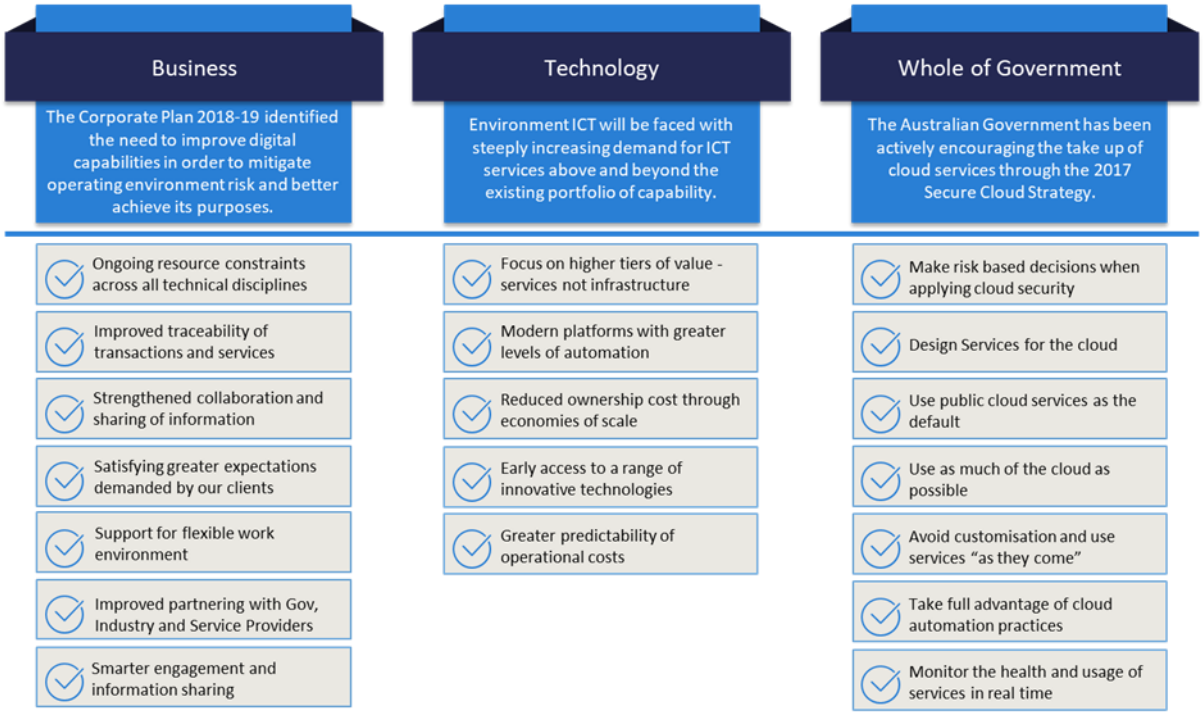

Figure 1 – Departmental Cloud Drivers

The absence of a strategic approach to cloud adoption may significantly increase the Department's ICT operational costs and exposure to security, financial and compliance risks. Section 4 of this document defines high-level strategic principles for the provision, operation and consumption of cloud services.

Section 5 of this document outlines how the Department will ensure that cloud services are adopted and managed in an efficient and well-governed manner, by pursuing the following four key initiatives:

- **ICT Operating Model Transformation:** The adoption of cloud services requires a new operating model focused on delivering ICT value through commodity services. Only core enterprise capabilities that cannot be delivered by third parties should be sustained on-premises.

- **Data and Information Management:** The Department will review and update existing data and information management policies and standards with a focus on ensuring the challenges and opportunities of the adoption of cloud can be appropriately addressed.

- **Operational Readiness:** As services are migrated to the cloud, core support services must also be evaluated and transformed to ensure they continue to satisfy operational obligations and demands.

- **Workforce Skills Alignment:** The Department will build capability in the general management, governance and control of cloud services, and invest in training technical staff in completely new skillsets for the Department, as well as a re-factoring of the existing skillset baseline.

Section 6 of this document details the Department's approach to cloud adoption. The Department will utilise a Hybrid Cloud Decision Framework to help guide the selection, consumption, placement, and operation of applications and ICT services, both using Department owned infrastructure and in the cloud. The framework represents an iterative approach intended to accelerate the assessment of services against a range of business and operational criteria. The framework will be used for both existing departmental applications and new services. Section 6.2 provides a high-level view of the key components of the proposed Hybrid Cloud Decision Framework. **Figure 2** shows the high-level cloud adoption and optimisation lifecycle for the Department.

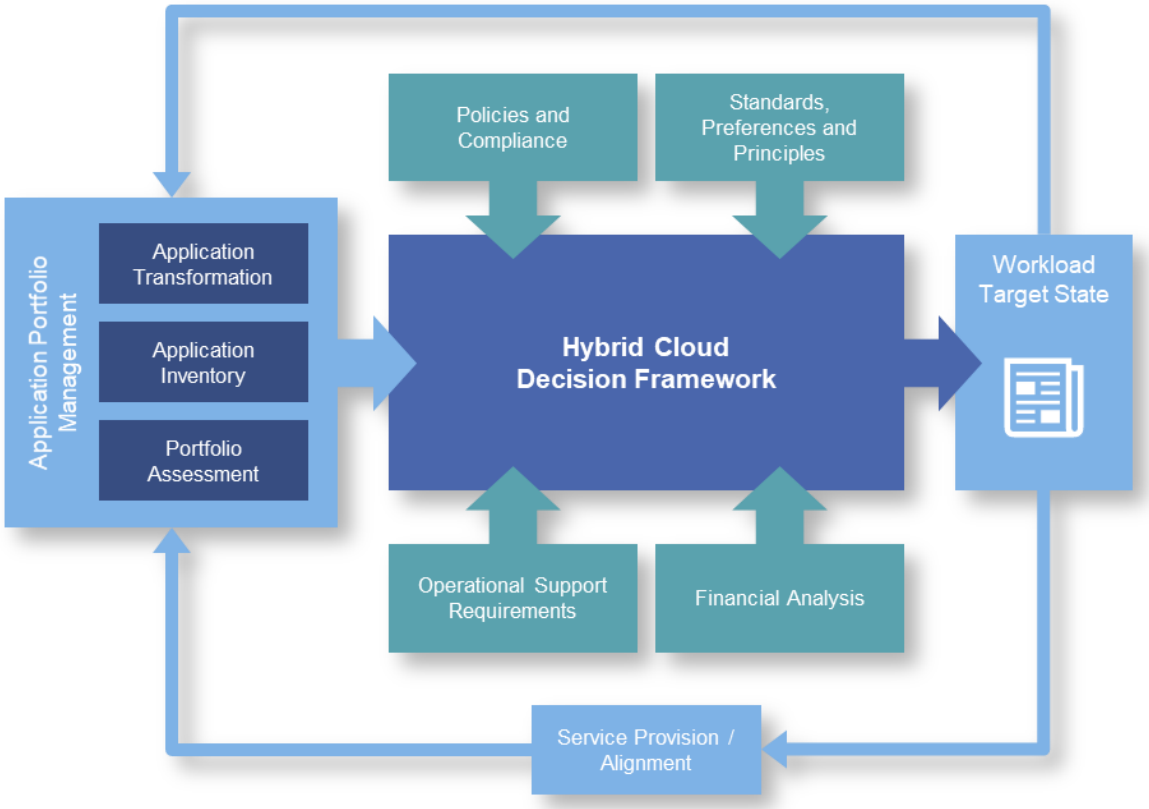

Figure 2 – High Level Cloud Adoption and Optimisation Lifecycle

Adoption of cloud services should prioritise off the shelf and configured services before considering customised options. Where customised services are required, Platform as a Service (PaaS) capabilities should be considered before Infrastructure as a Service (IaaS) or Department owned infrastructure. Core corporate systems should leverage several Software as a Service (SaaS) offerings. Where it is required, a range of these service offerings have been, or are on the path to be, certified to an Australian Government Protected level (Reference B).

The Department is also developing a Hosting Strategy, which will outline how adoption of cloud services will integrate with the Department's broader ICT architecture, and how the Department will select the most appropriate architecture for deployment of new and existing ICT solutions.

2 INTRODUCTION

2.1 What is Cloud - Definitions

'The cloud' means storing and accessing data and programs by way of the Internet. Cloud services deliver on-demand computing resources over the Internet as an alternative to maintaining Department owned physical ICT infrastructure, corporate data centres, platforms and software.

According to the official National Institute of Standards and Technology (NIST) definition[1], 'cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.'

Cloud is supported by a marketplace of vendors and suppliers who provide cloud computing capabilities, on public, multi tenanted or single tenanted platforms, where consumers can source computing resource, networks, servers, storage, software and applications as a service using consumption-based pricing.

The five essential characteristics of a cloud platform are generally accepted to be:

- **On-demand self-service:** A Department can provision computing capabilities, such as server resources, network storage, or application interfaces as needed automatically without requiring administrative interaction.
- **Network access:** The platform is available over the network and accessed through standard mechanisms that promote use by mobile phones, tablets, laptops, and workstations.
- **Resource pooling:** The provider's resources are pooled to serve multiple consumers using a multi-tenant model. Physical and virtual resources are dynamically assigned and reassigned according to agency demand. Examples of resources include compute, storage, and network.
- **Elasticity:** Capabilities can be elastically provisioned and released to scale rapidly outward and inward in line with demand. The capabilities available for provisioning services often appear to be unlimited and can be appropriated in a range of quantities and at any time.
- **Measured service:** Cloud systems automatically control and optimise resource use by applying a metering capability appropriate to the type of service (e.g. storage, compute, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and the Department.

Cloud computing can be delivered in many forms, the differences between the different forms are largely defined by the boundary between what the service provider manages and what the Department is responsible for. **Figure 3** compares the level of services provided by a cloud service provider under each form of cloud computing architecture to the traditional on-premises architecture. The diagram assumes that traditional ICT is fully managed in-house using internal capability.

[1] The NIST Definition of Cloud Computing (NIST Special Publication 800-145)

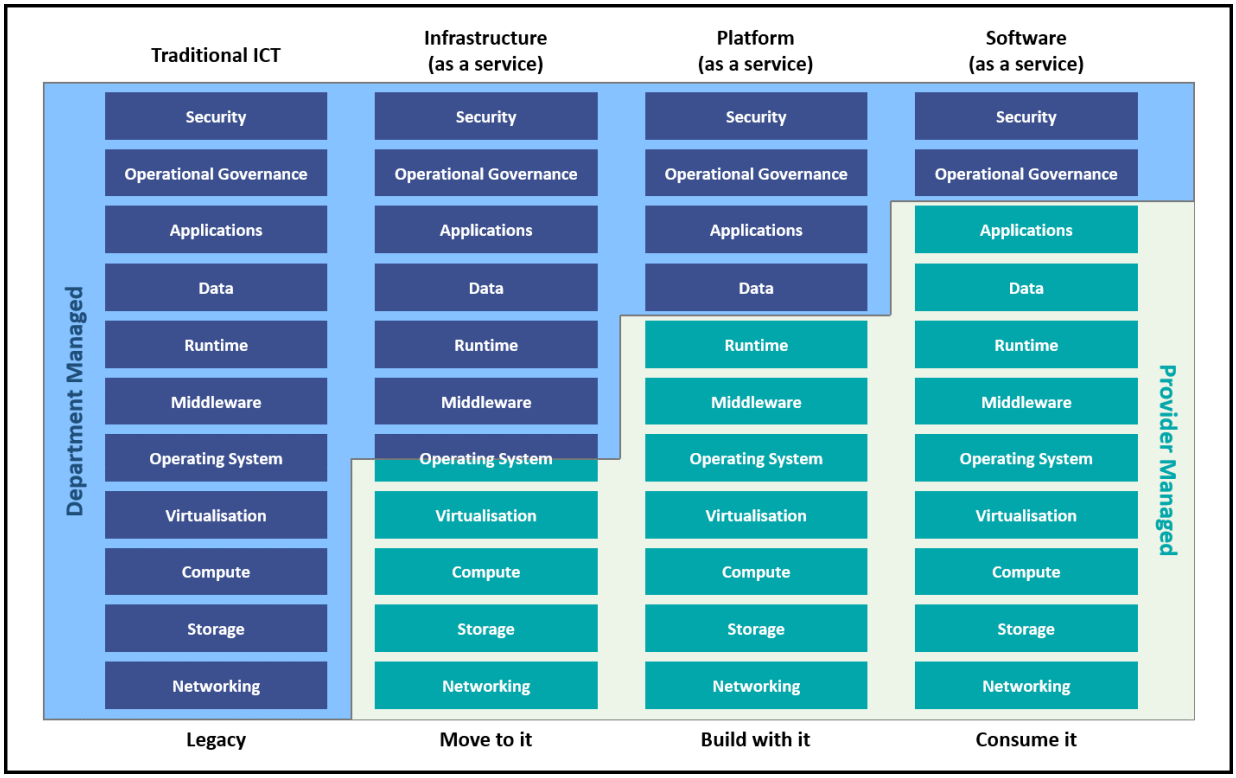

Figure 3 – Comparison of cloud computing architectures

These system delivery forms are defined as follows:

- **Traditional ICT:** This is the traditional delivery method of ICT services. In this model, all ICT services are owned, managed and operated by the Department or their outsourced service provider.

- **Infrastructure as a Service (IaaS):** In this model the service provider provisions physical hardware and data centre capability to the Department including compute, storage and networking. The service provider also manages any virtualisation and often provides preconfigured operating systems for deployment. The Department does not manage or control the underlying cloud infrastructure but has control over some of the system network, operating systems, storage, and the deployed applications.

- **Platform as a Service (PaaS):** This is a model where the service provider takes on a higher level of responsibility to deliver the Department a capability to build on. Along with the underlying cloud infrastructure, there are no operating systems for the Department to maintain but rather a set of cloud native platforms such as managed database or container services.

- **Software as a Service (SaaS):** In this model, all application capability is delivered fully 'as a service' with no traditional ICT services required to be operated by the Department. The Department is only responsible for application configuration, administration and subscription management.

The Department will use a combination of these four system delivery forms, with automatic configuration and deployment of services to ensure efficiency and portability. The implications of moving from traditional ICT delivery to cloud services are discussed in section 5.

The generally available deployment models are:

- **Private cloud:** The cloud infrastructure is provisioned for exclusive use by a single organisation. It may be owned, managed, and operated by the organisation, a third party, or some combination of them, and it may be hosted by the Department or by a third-party service provider.

- **Community cloud:** The cloud infrastructure is provisioned for the exclusive use of a specific community with a common set of requirements or concerns. For example, a number of Australian Government and Critical Infrastructure Community clouds exist with a focus on meeting the Australian Government's protected classification security requirements.

- **Public cloud:** The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organisation, or some combination of them. It exists on the premises of the cloud provider.

- **Hybrid cloud:** Hybrid cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardised or proprietary technology that enables data and application portability. However, can often be used to describe an environment comprising of both traditional, organisation owned, or private cloud in combination with Public or Community cloud.

The funding, consumption and ongoing operations of cloud-based services are different from the traditional means of systems delivery previously practiced by the Department. The adoption of cloud services will result in a shift from capital expenditure to operational expenditure. The terminology and definitions of these concepts are:

- **Capital Expenditure (CapEx):** The traditional funding approach for delivery of a capability and/or equipment. The Department spends these funds on fixed assets, such as the purchase, maintenance, and improvements of ICT equipment such as servers and storage arrays. Where costs are identified and funding provided up-front. Procurement costs show up on the Departments balance sheet, and the expenditure is subsequently depreciated over several years, in alignment with relevant tax regulations. As cloud-based services are not capital assets, the Department's adoption of cloud-based services should gradually reduce its CapEx on ICT over time. As the Department expands its mix of cloud technologies and service providers, opportunities to transition new services to the cloud should see a more noticeable drop in CapEx.

- **Operating Expenditure (OpEx):** These are the funds the Department uses to run its day-to-day business. OpEx items are generally consumed within the year they are purchased. OpEx purchases cover pay-as-you-go items that show up on the Department's profit and loss statement, and they are deducted from income as they occur. Cloud services being utility in nature are normally presented in an OpEx consumption arrangement. The Department's OpEx will increase commensurate with its adoption of cloud-based services.

The Department's adoption of cloud services also requires a shift in system delivery from development to integration:

- **Plan, Build, Run:** This is a typical system life cycle where user requirements are captured up-front and a system is built and maintained based on those requirements.

- **Plan, Integrate, Manage:** While the Plan, Build, Run model has worked well for the traditional ICT environment cloud presents an opportunity to rapidly deliver more generic offerings with simple light-touch configuration. The focus then becomes integration with the Department's operational systems and management of a multi-service provider arrangement.

2.2 Benefits of the Cloud

The Department recognises cloud services as key to supporting the delivery of business outcomes, now and into the future. The benefits of cloud adoption for the Department include the enablement of a more flexible and modern workplace, faster delivery of new Government programs, and initiatives and enhanced cross agency and community engagement.

Adoption of cloud services will allow the Department to leverage innovative new technologies. Through effective governance, the Department should achieve greater value from its investment in Information Technology.

The Department's business areas will benefit from faster deployment of ICT solutions with less upfront investment, enabling more efficient implementation of new programs and initiatives. Cloud services are typically on-demand meaning they can be enabled and disabled as required, allowing for greater control of costs.

Some benefits typically seen in other organisations which have adopted cloud services include the ability to collaborate at the same time on the same version of documents, regardless of geographic location. Furthermore, universal access to documents across a larger range of devices, not just traditional laptop and desktop devices, will provide the ability to enable and support flexible working arrangements.

The Department's stakeholders, including the general public, will benefit from standardised user interfaces that can be accessed across a variety of platforms and devices. The general public may also benefit from more rapid implementation of Government programs and initiatives which include an ICT component.

3 KEY DRIVERS

The Department recognises that the drive to consume cloud services will primarily be driven by the following:

- **Whole of Government Drivers:** the Australian Government's Digital Transformation Agenda encourages agencies to adopt cloud-based services where appropriate. Agencies are required to develop strategies for adoption of cloud-based services by early 2019.
- **Business Drivers:** the Department's *Corporate Plan 2018-19* identified the need to improve its technology and digital capabilities in order to mitigate the risks in its operating environment and better achieve its purposes.
- **Technology Drivers:** faced with steeply increasing demand for ICT services, the Department needs to rationalise and modernise its ICT to optimise use of shrinking resources and better support engagement with stakeholders.

3.1 Whole of Government Drivers

The Australian Government has been actively encouraging the take up of cloud services for a number of years and the *2017 Australian Government Secure Cloud Strategy* (Reference A) has introduced seven Cloud Principles to guide and drive agencies' adoption of cloud services. It encourages agencies to maximise the value of cloud to their business by using the cloud principles as a starting point to set their own vision and strategy for cloud adoption and requires these agency strategies to be complete by early 2019. **Table 2** outlines the Secure Cloud Strategy's Whole of Government guidance for adoption of cloud-based services.

Principle	Guidance
Make risk based decisions when applying cloud security	Risk based decisions, rather than just checking off compliance, are required to understand the security needs of a cloud service and apply the appropriate security controls.
Design Services for the cloud	Cloud techniques increase the speed at which resources can be accessed and used, reduce the manual tasks through automation and allow applications to be run independent of the infrastructure, enabling more opportunities for provision and expansion of services.
Use public cloud services as the default	The public cloud market offers a broad range of services and providers that enable agencies to keep their technologies and business processes up to date. Public cloud can provide fast and competitive options for agencies.
Use as much of the cloud as possible	Agility comes from models that leverage standardised cloud technologies. This enables agencies to keep pace with industry disruptions and innovation cycles as well as maintaining business process and technology currency.
Avoid customisation and use services 'as they come'	Agility comes from using the service 'as it comes' without bespoke processes being introduced which erode the business agility of the service by adding complexity and requiring intervention during change cycles.
Take full advantage of cloud automation practices	Automation enables support teams to focus on the more complex requirements that are unique to their business by minimising the effort need to provision, configure, backup, restore, patch, update and deploy services.
Monitor the health and usage of services in real time	Monitoring allows agencies to have visibility of their cloud usage, cloud health and enable them to control costs.

Table 2 – Whole of Government Drivers

3.2 Department Business Drivers

The Department's adoption of cloud services and technologies is in response to the rapidly evolving operational environment.

To enable the organisation to deliver more effectively each of these outcomes, the Department's corporate plan identified eight key areas of focus:

1. Preparing our workforce for the future
2. Partnering for better outcomes
3. Maintaining a positive risk culture
4. Making evidence-based decisions and providing evidence-based advice
5. Fostering innovation
6. Improving our technology and digital capabilities
7. Communicating and engaging effectively
8. Pursuing regulatory maturity

The efficient delivery of supporting ICT services, together with the innovation cloud services can provide, will be critical to the Department successfully delivering against these desired outcomes.

Table 3 outlines the Department's business drivers for adoption of cloud-based services.

Drivers	Cloud Value	Value Attributes
Greater sharing of responsibility and costs	Improved traceability of transactions and service boundaries/exchange.	Well defined interfaces. Metered visibility of costs and (potential) charge-back options. Business process mapping and supporting analytics.
Strengthened collaboration	Reduced investment to gain access to improved tools for document management, analysis and sharing of data with accompanying security controls.	A wide range of advanced tools for search, analysis and distribution of information and data. Relatively small investment gains considerable capability. Many on premise products have equivalent hosting options in the cloud if required.
Greater client expectations	Rich customer interfaces across a wide range of channels can be employed with few barriers to adoption.	Customer interfaces employ standards across different channels that simplify maintenance and content delivery. Standard content platforms mean that information can be maintained in one place yet accessed through rich, engaging interfaces. Opportunities to use applications and devices that are very familiar to the Department's clients.
Ongoing resource constraints	Reduces the requirement to sustain skills and staff in all aspects of technology stacks. Opportunities exist to specialise in technologies that are of immediate value to the department and consume services that are provided better, cheaper and faster by other parties employing cloud technologies.	Opportunities to reduce sustainment costs due to the economies of scale obtainable from cloud technologies and service providers. Pathways to reduce technical debt from legacy systems through rationalisation and modernisation. Automatic technology updates mean that the cloud vendor deals with the update of the technology and system. Options exist to deliver more with static or declining resource profiles employing cloud technologies, development and deployment approaches.

Table 3 – Business drivers

3.3 Technology Drivers

The Department's ICT area will be faced with steeply increasing demand for ICT services beyond the support of existing systems. Demand for new services and associated innovative solutions to meet emerging business requirements will almost certainly increase. **Table 4** summarises the Department's ICT operations key drivers for the adoption of cloud services.

Drivers	Cloud Value	Value Attributes
Resource optimisation	Focus on higher tiers of value chain, services not infrastructure.	Minimisation of capital assets requiring maintenance resources. More effort focussed on business facing capabilities.
Rationalise and Modernise	Modern platform alternatives with highly evolved migration pathways and automation.	Reduction in the need to 'start from scratch' when considering modernisation options. New capabilities such as automation, metering, reporting and reduced maintenance costs are ongoing benefits that can be leveraged to further reduce overheads.
Accelerated Remediation and Consolidation	The cloud provides a wide range of options to: • improve resilience • enhance recovery • add and remove capacity • improve security posture in line with Whole-of-Government requirements.	Reduced total cost of ownership across different cloud models due to economies of scale. New capabilities such as automation, metering, reporting and reduced maintenance costs are ongoing benefits that can be leveraged to further reduce overheads.
Improved Partnering with Government, Industry and Service Providers	Whole-of-Government services at levels of security that are compliant with Government requirements. A wide range of options for engagement with Industry and the public.	Well defined interfaces in a shared environment that is fully supported. Standard approaches to security and sharing data and undertaking transactions. Agile and efficient processes and development technologies to promote information exchange and collaboration.
Smarter Engagement	Modern toolsets and processes that greatly enhance the exchange of information and work practices between ICT and business subject matter experts (SME).	Information Sharing across a varied team. Transparency of process. Responsiveness and Availability. End to end engagement from early idea generation through to realisation.

Drivers	Cloud Value	Value Attributes
Enabling Innovation	Early access and accelerated development of a range of innovative technologies of which machine learning, and blockchain are better known examples, but also Internet of Things (IoT) processing and materials exploitation (for example, remote sensing/drone imagery.	Experimentation and agile/incremental development of capabilities without extensive upfront investment. Opportunities in technologies, adoption and migration strategies. Flexibility supporting rapid and iterative change. Implementation of many indirect, but important technologies such as security, data exchange and collaboration. High degrees of effective and efficient automation.
More predictable costs	Opportunity to consume services which can be provided better, cheaper and faster than on-premises solutions.	Negates need for expensive and hard to find specialist skills to develop and maintain on-premises solutions.

Table 4 – ICT Strategic drivers

4 CLOUD PRINCIPLES

The purpose of this section is to define the high-level strategic vision for the provision, operation and consumption of cloud services by the Department. It highlights the key principles that the Department aspires to achieve once this strategy has been implemented:

1. **Consider Cloud First:** Maximise the consumption of cloud services where they deliver real business value and can be consumed in a compliant manner.

2. **Cloud Choice:** Provide lines of business with choice and flexibility in the selection and consumption of cloud services to ensure that business needs are met.

3. **Rationalise and Standardise:** Assists with successful cloud adoption, limits operational impact and provided financial benefits to the Department.

4. **Seamless and Efficient Operations:** Harmonise both on-premises and off-premises service management and support operations.

5. **Secured and Governed Consumption:** Consume cloud services in a secure, consistent and compliant manner without complexity or delay.

6. **Modernised Datacentre:** Bring cloud like efficiency and functionality to on-premises operations through modernisation and cloud integration.

7. **Ease of Consumption:** Accelerate business access to cloud innovation though user driven self-service consumption, within the preapproved service offerings.

8. **Business Process Alignment:** Transform business processes to best leverage the strengths of cloud offerings whilst addressing identified weaknesses.

4.1 Consider Cloud First

The public cloud market offers a broad range of services and providers that enable organisations to keep their technologies and business processes up to date.

To help drive the potential benefits of cloud into government organisations, the Digital Transformation Agency (DTA) has advised that agencies should consider cloud when adopting new or updating existing services. Specifically, the DTA has requested agencies consider public cloud in preference to any other cloud deployment model.

This statement from DTA is not a directive to utilise cloud in every case. Instead, it is a directive that agencies consider the use of cloud services as a first consideration, not a mandate. In all cases, cloud services should only be adopted where they are right for the organisation and have been assessed across a range of factors, including business and financial value.

Further, regardless of the power of any cloud service, their provision and use must still comply with regulatory requirements, such as those related to security and the appropriate treatment of government information.

In alignment with the Australian Government's vision for the use of cloud services, the Department will implement a 'consider cloud first' approach when adopting or updating ICT based services.

The Department will look to develop policies, supporting technologies and decision frameworks to support cloud service evaluation. These mechanisms will accelerate the assessment, development or selection of cloud services whilst at the same time ensuring that they can be consumed in cloud with sufficient benefit to the organisation and in a fully compliant manner.

4.2 Cloud Choice

The number of business and technical services available as cloud offerings is significant and continues to grow. These offerings currently include thousands of SaaS applications, a diverse range of PaaS platforms and a growing number of public and community IaaS providers.

Industry analysts predict that most enterprise organisations will likely consume services across several IaaS and PaaS platforms and tens of SaaS applications by 2020. Rapid and well-managed access to these cloud services can help organisations realise innovation opportunities faster and with less effort. Further, by providing managed access to a wide range of cloud services, organisations can help avoid or reduce unmanaged shadow IT consumption.

To ensure that business needs are met, the Department will support a multi-cloud, multi-provider approach through all aspects of the business and ICT operations.

Seeking to maximise cloud choice does not mean that cloud services can be consumed without appropriate guidance and governance. To ensure cloud services are selected, integrated and consumed in a controlled and well-governed manner, the Department will agree and instigate a range of governance controls and capabilities. This will include the establishment of pre-approving providers/services, enforcing of standards, and having the right tools in place to manage multi-cloud.

In addition, quality advice and guidance around the selection of cloud services will be the result of establishing and maintaining a skilled workforce. Adopting an open standard for architecture will further support multi-provider utilisation and prevent vendor lock-in. On-boarding and off-boarding arrangements will form part of contractual arrangements with cloud service providers.

4.3 Rationalise and Standardise

The Department's move to cloud provides an opportunity to evaluate and better understand its application landscape. Application rationalisation, modernisation and standardisation is an integral activity of the adoption of cloud technologies. This will be critical as it is highly likely that there will be two classes of environments, the cloud environments and the legacy/on-premises environment. A managed application rationalisation exercise is a critical part of planning cloud technology adoption. The rationalisation exercise should include the identification and decommissioning of applications, consolidation of redundant applications and partially or completely refactoring. This exercise will also assist in identifying applications to be left alone.

Adopting cloud technologies generally requires converting vertical stacks into horizontals comprising of infrastructure, platform and enterprise software. The move to horizontal stacks is possible if proper hardware and software standards are adopted. Hardware and software standardisation is a critical driver for most of the cloud benefits and requires balancing act between operational priorities and transformational aspirations. Standardisation assists with scaling the operational impact and can provide financial benefits to the Department.

Modernisation follows rationalisation and standardisation as applications that are in the cloud or transitioning to cloud technologies need to measure up to the standards that have been set. The following are examples of what modernisation standards should include:

- standard development languages
- central administration of security policies
- refactoring existing applications.

4.4 Seamless and Efficient Operations

Effective hybrid cloud management reduces complexity and management costs inherent with multiple environment coordination. The Department will benefit by being able to focus more on delivering business services and ensuring the full benefits of cloud are realised.

A unified operating model for both on-premises and cloud services will include common architectures, technologies, processes and policies. Full alignment with this operating model can bring cloud-like capabilities to on-premises datacentre.

4.5 Secure and Governed Consumption

Cloud technologies bring with it concerns and uncertainty in regards to security and operational surety. Establishing the capability to quickly and easily achieve compliance will accelerate and ease the adoption of cloud, whilst also providing confidence that the Department is operating in a consistent and secure manner.

Building a flexible yet compliant supporting framework will allow for secure and well-governed consumption of cloud services. The framework will be built on clearly defined and understood policies and processes combined with automated security and governance capabilities. The framework will align with and support the Department's cyber security strategy and policy.

4.6 Modernised Datacentre

The Department will have Department owned and self-hosted facilities for an extended period. By modernising this environment and transitioning to low touch operational approaches, efficiencies and uniformity with cloud services can be achieved. Modern approaches will provide out of the box capabilities, automation, self-configuration, and extensibility. Department owned infrastructure needs to be made cloud ready with native integration with cloud services. This will ease and accelerate migrations and integration to and from the cloud.

4.7 Ease of consumption

Ease of consumption will dramatically improve the speed of delivery and reduce reliance on IT. The use of self-service catalogues of pre-approved services supports and allows for the use of push of button provisioning of workloads.

4.8 Business Process Alignment

To ensure efficient adoption of cloud services, reduction in complexity, and predictable outcomes, business processes must also be transformed. The greatest benefits will be achieved by evaluating and implementing the changes required from ICT operations right through to line of business operations. Upskilling in cloud technologies across all areas of the business will allow staff to reassess and refocus existing processes to maximise the potential value of cloud services.

5 CLOUD FOUNDATIONS

As the Department increasingly adopts cloud services, ICT and business operations must be assessed and potentially transformed. This is to ensure that cloud services will be both adopted and managed in an efficient and well-governed manner. Changes will be required across many aspects of the Department's business operations, and include re-consideration of people, process, information and technology.

The Department has identified four key Initiatives that will be pursued to support the adoption and management of cloud services, each of which are described in further detail in the following sections:

- ICT Operating Model Transformation
- Data and Information Management
- Operational Readiness
- Workforce Skills Alignment

5.1 ICT Operating Model Transformation

The role of the ICT teams will transform from being a builder of ICT capability to a broker of business services. This shift allows ICT to satisfy business demand for innovation, address the growing momentum from government relating to digital transformation, while ensuring appropriate controls and governance remain intact.

Cloud technologies provide the Department the opportunity to significantly improve scalability, agility, and manage cost. Navigating the large number of competing and highly variable cloud providers and services can be complex. New services must be consumed, integrated and managed with appropriate levels of governance applied to ensure compliance with operational and business requirements, while establishing sufficient risk control. ICT must become a bridge between external providers and line-of-business operations to maximise business outcomes.

As the Department increasingly adopts cloud services, a different operating model is required. New architectures and strategic sourcing models must be established to harmonise transformation initiatives. The operating model will need to focus on delivering ICT value through commodity services, sustaining only core enterprise capabilities that cannot be delivered by third parties. The Department will pursue improved collaboration and smarter, more agile, engagements with cloud service providers in preference to the building of capital assets.

A transformation of processes, people and technology is essential. Toolsets, data, security, policies, financial control and user experience will now span multiple provider environments, both public and private. Service management must be extended to match this new landscape.

Service Integration and Management (SIAM)

Industry has recognised that Information Technology Infrastructure Library (ITIL) based ICT management models are not optimised to comprehensively support the demands of multi-provider hybrid cloud environments. Management of multiple vendors, and the orchestration of a common approach across different clouds, is essential for effective transformation, control and cohesion.

It is envisaged that the Department will eventually consume a wide range of cloud services, across multiple vendors. The Service Integration and Management model was developed to compliment ITIL and solve the challenges around following this path.

Service Integration and Management ensures that both business services and technology services are integrated to provide a single business-facing ICT organisation. The Department must therefore adopt Service Integration and Management to seamlessly integrate interdependent services from various internal and external service

providers into end-to-end services that meet business requirements. Service Integration and Management is a key determinant of the Department's success of its cloud adoption.

The adoption of Service Integration and Management will require the implementation of the following:

- **Cloud Policies, Procedures and Standards:** undertake a review of existing policies, procedures and standards to determine their applicability to the consumption and operation of cloud services. Policies, procedures and standards will be developed, revised or refined in accordance with the findings of the review.

- **Dedicated Service Integration and Management and Cloud Management toolsets:** implementation of dedicated tooling to support cloud management. For simplicity, the Department's intent is to operate with the minimum number of tools necessary to satisfy requirements. As such, the Department will prioritise the implementation of a Cloud Management Platform, with a suite of targeted solutions implemented to cater for any remaining gaps.

- **Financial Management:** examples across industry and government have shown that without commensurate transformation of financial arrangements, the adoption and operation of cloud services will be greatly inhibited. Specific areas of strategic focus include:

  - transition from capital to operational expenditure
  - improvements across cloud financial monitoring
  - management and ownership
  - establishment of a showback/chargeback process
  - improved resource on/off-boarding and subscription management.

- **Contract and Vendor Management:** uplift to the Department's procurement, contract and performance management capabilities to improve management and governance of cloud services and providers. This will be conducted through targeted training and resource hire.

5.2 Data and Information Management

The Department's current suite of data and information management policies and guidance, underlying operational structure and the resource skills of staff have been built around the existing on-premises operations. As the Department's cloud presence becomes more pervasive within its business areas, a lag in how current data and information management practices respond to potentially different requirements of a cloud environment may represent a challenge and potential risk for the Department to address. Disaggregation of critical departmental data assets across multiple providers and locations will increase the complexity of integration across services, may impede the Department's ability to coalesce and extract key information and insight, and could affect the Department's ability to satisfy regulatory and compliance obligations, such as privacy and records management. However, many of these challenges already exist under current on-premises arrangements. When moving data to the cloud, aspects such as security, access and cost should be considered in relation to business needs. As the Department reviews and updates existing data and information management policies, guidance and standards, the challenges and opportunities of the adoption of cloud will be addressed.

Table 5 outlines some key capabilities that will need to be reviewed to address the challenges and opportunities of cloud adoption.

Capability	Future Direction
Information Governance	The Department's information governance committees will be responsible for extending the Department's existing governance framework to implement cloud specific controls.
Information and Data Management Policies and Guidance	The Department's suite of information and data management policies and guidance will need to be reviewed to ensure that they support the Department's approach to the adoption of cloud technologies. In particular, the Policy will need to articulate what data can/cannot be stored in the cloud, the impact of Government directions for data availability and use; and the responsibility of information stewards with respect to data stored across different cloud services.
Data Quality and Remediation	End-to-end data quality management service to cloud-based information repositories including reporting and visibility of data quality across the multi-cloud.
Reference and Master Data	Planning and development activities to include cloud-based information and data in the context of the Department-wide approach to reference and master data management.
Metadata Management	Support will be required for lineage and traceability of data to source systems that are provided by cloud technologies and services; and, the visibility of this in a consistent way across all of the Department's systems.
Business Intelligence and Analytics	Expand current planning and development activities to provide for the consistent employment and sustainment of multiple cloud-based analytics systems with respect to data location, aggregation and accessibility.
Information Architecture	Develop, validate and extend the information architecture across cloud-based technologies and services in line with the Department's information requirements.
Data design, modelling and interoperability	In concert with the department's existing enterprise data model (EDM) development, promote a domain driven design (DDD) approach to ensure that the requirements and capabilities of cloud-based technologies and services are implemented, yet reconciled with the EDM such that integration supports information flows amongst the department's systems.
Data Publication	Update existing publishing processes, standards, guides and templates to support external availability of the Department's publications from cloud-based services and repositories.
Data Access and Security	Update the Department's data access and security processes and standards for the sharing of data (including sensitive information) from cloud-based services and repositories where appropriate.
Documents, Content and Records	Review the current departmental document, records and content management solutions for cloud readiness in line with the Department's program for digital record keeping. Review and update the Department's Records policy and strategy to support the use of cloud technologies and their potential impact. Other document and records systems need to be considered in this context. For these systems, strategies need to be developed or updated for their potential migration to cloud-based technologies.

Table 5 – Key capabilities for review

5.3 Operational Service Readiness

Cloud technologies represents a significant opportunity for the Department to drive innovation throughout its business and uplift technical resources to higher value tasks that maximise service efficiency and quality. Without appropriate governance mechanisms, core support services will fail to realise the benefits presented by cloud technologies.

In isolation, individual applications may drive specific enterprise platforms and architectures that are incompatible with existing departmental services or required cloud services in the future. Critical services may be deployed in unstable or unsupportable environments, affecting availability and the Department's external reputation. In addition, this may result in greater operational overheads, increased costs, greater resourcing burdens and broader skillset requirements, while affecting the Department's ability to meet business, departmental or Whole-of-Government compliance obligations.

The Department recognises that as services are migrated to the cloud, it is imperative that the core support services are also evaluated and transformed, as needed, to ensure that they can continue to satisfy the organisation's operational obligations and demands.

The Department has developed a number of conceptual hybrid cloud management architectures that define the operational services that will be required to support ICT management of cloud SaaS and hybrid cloud IaaS services, provided in **Appendix A** – IaaS Management conceptual Architecture and **Appendix B** – SaaS Management conceptual Architecture.

Table 6 describes the foundation operational services that will be targeted as part of the Department's strategic approach to enabling and sustainably supporting cloud service operations.

Capability	Future Direction
Identity Management	Departmental authentication standards defined, agreed and documented. A single federated identity capability enabling secure, flexible and extensible authentication and authorisation services to applications regardless of hosted location.
Backup	Hybrid cloud capable backup solution(s), which meet the Department's data protection requirements. Comprehensive support across on-premises, IaaS, PaaS and SaaS services.
Capacity and Performance Monitoring	Hybrid cloud capable monitoring and management solution(s) deployed and operational providing performance, capacity and enhanced situational awareness across on-premises, IaaS, PaaS and SaaS services. Integration with Departmental Service Management platforms and processes.
Archive	Departmental archive capability providing secure, cost effective and multi-use information retention services to applications and services on-premises and cloud. Support for industry standard protocols to avoid vender lock-in and orphaned data. Simplified data discovery, portability and information sharing. In line with the Department's archive policy and supporting the departmental archive compliance requirements.
Disaster Recovery	Disaster recovery solution(s) enabling the restoration of selected ICT enabled services within agreed departmental operational objectives, including the Department's BCP. Comprehensive support for business-critical applications and data, selective support for lower tier applications and data.
Cloud Security	A comprehensive, multi-layered hybrid cloud security capability delivering threat protection and control over the Department's sensitive data and systems regardless of user, device or service location.
Data Integration	Departmental data integration standards defined, agreed to and documented. Supporting API driven solution(s) deployed to enable the governed presentation, consumption and movement of information between departmental applications and services regardless of hosting location or provider.
Configuration Management	Configuration management solution(s) providing operational and configuration consistency across departmental systems and services regardless of hosted location/provider.
Brokerage and Orchestration	Comprehensive cloud management solution(s) providing simplified and flexible deployment of hybrid cloud services regardless of hosted location/provider. Integration with preferred providers, platforms and technologies to enable rapid development and deployment with operational consistency.
Hybrid Cloud Financial Management	Hybrid cloud financial management capability enabling accurate cost estimation for the deployment of services both on-premises and in the cloud, including: • cost estimation and quoting • through life financial monitoring • accountability via charge back/show back • invoicing management.

Table 6 – Foundation operations services

5.4 Workforce Skills Alignment

Beyond training the Department's workforce to uplift capability relating to the general management, governance and control of cloud services, the Department will also invest in training technical staff across a targeted range of digital skills. These include completely new skillsets for the Department, as well as a re-factoring of the existing skillset baseline.

Without this skills transition, the organisation will not be able to fully leverage the benefits of cloud, resulting in a restriction of the offerings that can be leveraged from cloud technologies, slower time to production, less likelihood of realising cloud benefits, and greater risk relating to transformation of services.

Although not exhaustive, **Table 7** outlines the skills that the Department will initially target for training and resource uplift.

Skillset	Development Needs
Development and Coding	Application developers will need to build skills in technologies and approaches for the development of cloud native applications.
Cloud Architecture	Architects will need to develop greater knowledge in cloud architectures, practices that improve alignment with emerging DevOps models.
Cloud Security	Security skills need to adapt to establish knowledge of the security practices, risks and mitigations for cloud environments, including application security practices.
Automation	Traditional infrastructure provisioning will need to evolve to focus on infrastructure as code, developing skills in blueprinting for provisioning and enhancing the understanding of cloud network topologies.
Infrastructure Support	Infrastructure support staff will need to build skills in cloud services and improve skill in business analysis.
Service Management	Service managers will need to develop an end-to-end understanding of cloud services in order to manage contracts with cloud service providers across different forms of cloud computing.
Support	Helpdesk staff will need to build skills in cloud environments and cloud applications.

Table 7 – Skills alignment

6 CLOUD ADOPTION

The Department will initially focus its cloud adoption efforts on the following priorities:

- Establish cloud-based services that directly support business objectives, where usage of cloud services makes sense (for example, using public and private cloud capabilities to support and drive the adoption of the digital workplace strategy and roadmap to the department's staff).
- Modernise and rationalise the Department's business systems to align with cloud native architectures, where the business process and the system supports the adoption of cloud technologies, in preparation for transition to cloud services.
- Review of the Department's business processes for compatibility with the adoption of cloud services.
- Focus on cloud technologies which minimise the Department's costs and provide maximum business level benefits.
- The introduction of agile, automated and on-demand capabilities and accompanying work practices.

The Department's approach towards cloud adoption has been developed in alignment with the Whole of Government 'consider cloud first' policy. To support this adoption, the Department will use the Hybrid Cloud Decision Framework at **Figure 2** to guide the consumption, placement, and operation of applications and ICT services, both on-premises and in the cloud.

The framework represents an iterative approach intended to accelerate the assessment of services against a range of business and operational criteria. Whether an existing Departmental application, or a new service, the Department will leverage this framework to guide the development, selection, placement and optimisation of application services.

The framework will be further developed, maintained and administered by the Department's architecture authorities, ensuring integration with existing architectural governance and control.

The following sections outline the key components of the Department's Hybrid Cloud Decision Framework.

6.1 Application Portfolio Management

The Department recognises that a deep understanding of their application environment is vital to the successful adoption of cloud services.

To ensure that the Department has the necessary understanding it will seek to assess and where necessary, augment their existing application management practices. Specifically, any review will look to ensure information critical to the assessment and management of applications for cloud consumption is present, accurate and well maintained by these practices.

Whilst a fully developed Application Portfolio Management capability provides a wide range of services, the following specific functions have been identified as critical to the adoption of cloud services:

- **Application Inventory:** Maintain an inventory of applications across the organisation including key application characteristics.
- **Application Portfolio Assessment:** Conduct comprehensive assessments against the application inventory and recommend application transformation, retirement, replacement and other treatments as necessary.
- **Application Transformation Roadmap:** Develop and maintain a roadmap to guide the development and transformation of the application within the portfolio.

6.2 Hybrid Cloud Decision Framework

The Hybrid cloud decision framework represents the establishment of a new capability for the Department. This will require the development of a suite of new processes, implementation of tooling, as well as additional or refocussed resourcing to drive adoption of cloud in a consistent, efficient and considered manner. Applications and services will be evaluated on a cyclical basis, identifying opportunities to drive greater efficiency for the Department through transformation, provisioning and service alignment activities. This will rapidly drive the Department's cloud agenda while maintaining the integrity of systems and adhering to departmental controls and governance.

Actual guiding criteria will be determined during the development of the framework however; the key criteria may include, but are not limited to:

- cloud policy
- application strategies and portfolio
- compliance and regulatory obligations
- technical and architectural standards, principles and guidelines
- security
- business requirements
- operational support requirements
- value for money and financial characteristics
- consumption preference order (i.e. SaaS, PaaS, commercial off-the-shelf, configurable, and customised).

Once an application service has been evaluated through the decision framework, the resultant output will be an application Workload Assessment. This Workload Assessment will provide a consistent view of the applications alignment against business and operational requirements.

Based on this workload assessment, business can make an informed decision concerning the ongoing consumption, provision, hosting management of the application services.

The following services are considered prime candidates for transition to the cloud:

- customer-facing applications (for example, customer portals)
- online training, forms and documentation (including guides and reference material)
- integration with partner or external customer applications and cloud-based services
- online booking and payment gateways (e-commerce)
- cloud management and service brokerage systems.

The use of Cloud management and service brokerage systems will be key to ensuring the successful adoption and implementation of the governance framework, as well as enabling the correct information is provided for any decisions. The cloud management platform should support traditional/departmental owned infrastructure, as there is likely to be requirements for this at least in the short to medium term.

6.3 Application Transformation

The Department recognises that the ability for the organisation to leverage cloud services, particularly PaaS and IaaS, is intrinsically linked to the organisation's application strategy. **Table 8** outlines the application transformation options available to the Department.

Future	Description
Re-host	Redeploy applications to a different hardware environment and change the application's infrastructure configuration. Re-hosting an application without making changes to its architecture can provide a fast cloud migration solution. However, the primary advantage of IaaS, that—teams can migrate systems quickly, without modifying their architecture—can be its primary disadvantage as benefits from the cloud characteristics of the infrastructure, such as scalability, will be missed. This method is normally only used for short-term solutions as long term it is generally more expensive than leaving the application on organisation owned infrastructure.
Re-factor	Run applications on a cloud provider's infrastructure. The primary advantage is blending familiarity with innovation as 'backward-compatible' PaaS means developers can reuse languages, frameworks, and containers they have invested in, thus leveraging code the organisation considers strategic. Disadvantages include missing capabilities, transitive risk, and framework lock-in. At this early stage in the PaaS market, some of the capability's which developers depend on with existing platforms can be missing from the current PaaS offerings.
Revise	Modify or extend the existing code base to support legacy modernisation requirements, then use re-host or re-factor options to deploy to cloud. This option allows organisations to optimise the application to leverage the cloud characteristics of providers' infrastructure. The downside is that kicking off a (possibly major) development project will require upfront expenses to mobilise a development team. Depending on the scale of the revision, revise is the option likely to take most time to deliver its capabilities.
Rebuild	Rebuild the solution on PaaS, discard code for an existing application and redesign the application. Although rebuilding requires losing the familiarity of existing code and frameworks, the advantage of rebuilding an application is access to innovative features in the provider's platform. They improve developer productivity, such as tools that allow application templates and data models to be customised, metadata-driven engines, and communities that supply pre-built components. However, lock-in is the primary disadvantage so if the provider makes a pricing or technical change that the consumer cannot accept, breaches service level agreements (SLA), or fails, the consumer is forced to switch, potentially abandoning some or all of its application assets.
Replace	Replace; for example, discard an existing application (or set of applications) and use commercial software delivered 'as a service'. This option avoids investment in mobilising a development team when requirements for a business function change quickly. Disadvantages can include inconsistent data semantics, data access issues, and vendor lock-in.

Table 8 – Application transformation types

APPENDIX A – IAAS MANAGEMENT CONCEPTUAL ARCHITECTURE

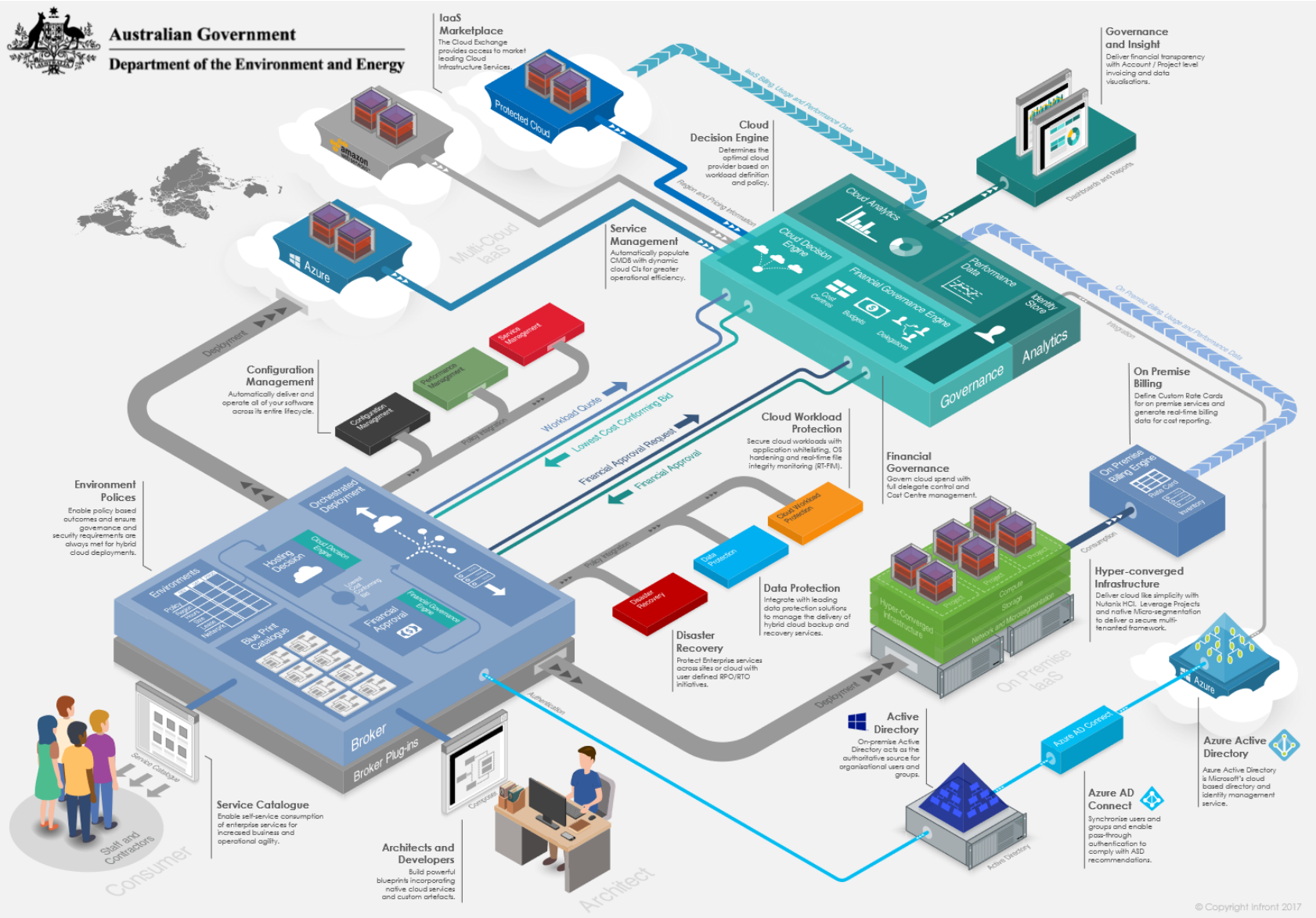

Figure 4 – IaaS management architecture

APPENDIX B – SAAS MANAGEMENT CONCEPTUAL ARCHITECTURE

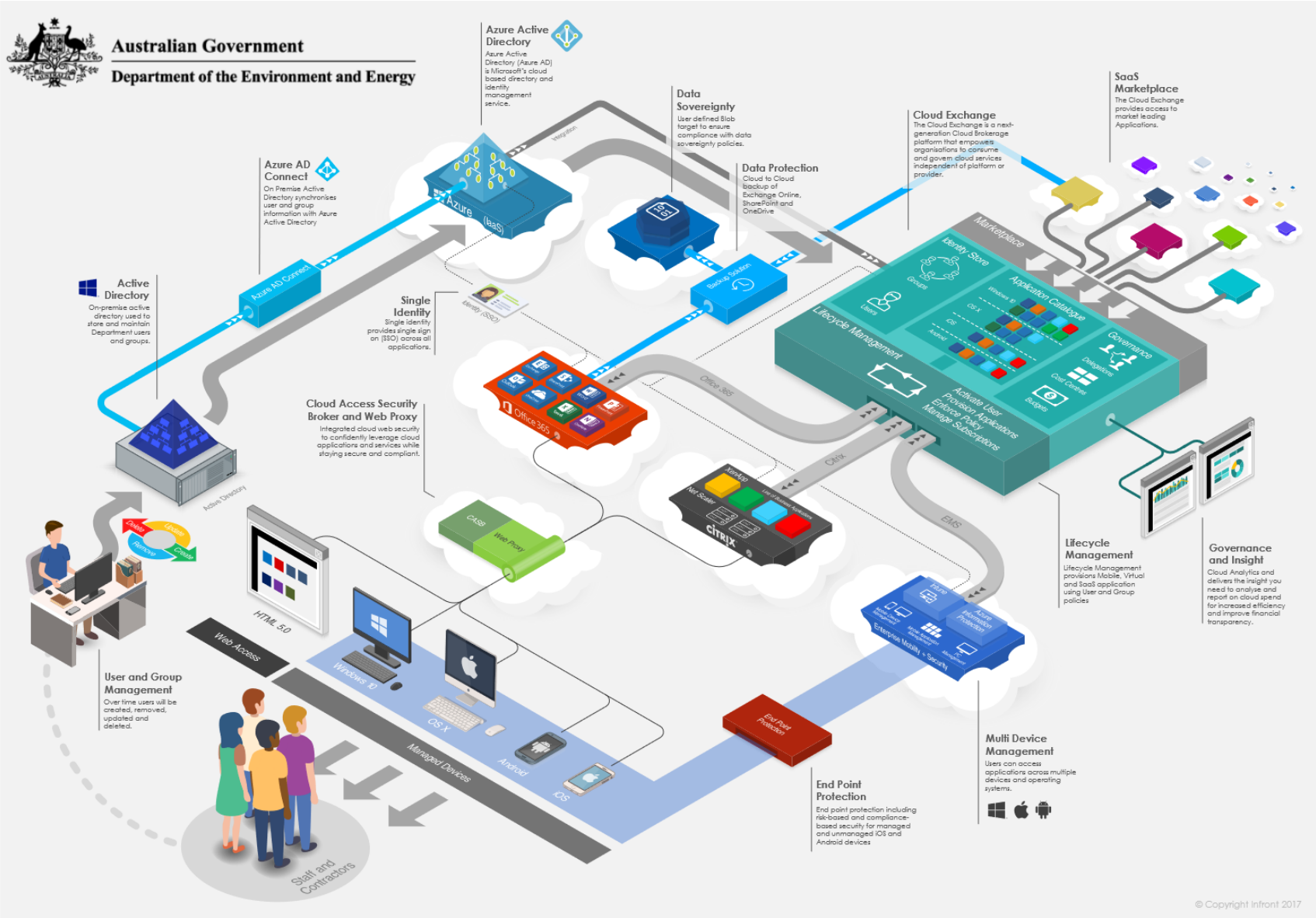

Figure 5 – SaaS management architecture